

DATA
INTELLIGENCE
READINESS
MISSION SUCCESS

SentinelAI

AI-Powered Predictive Maintenance
for Mission Readiness



PREDICT

Identify risks early



PREVENT

Reduce downtime



PROTECT

Ensure mission success

A SMARTER TOMORROW
FOR CRITICAL ASSETS



Team Name
NexGen



Team Lead
Aayush Savaliya



Team Members
Jeel Pipaliya | Krish Singh | Jevil Savani



University
CHARUSAT UNIVERSITY
AIML Department

BUILT FOR A SAFER,
STRONGER TOMORROW

THE PROBLEM

Mission Readiness **at Risk**

Critical assets can fail unexpectedly, while valuable data remains under-utilised.

**WHAT****Unexpected Asset Failures**

- Critical components can fail without warning
- Maintenance relies on fixed schedules
- Early warning signs are often missed in complex data

**WHY****It Hurts**

- Mission delays and operational downtime
- Higher maintenance and recovery costs
- Increased risk to equipment and personnel

**WHO****Is Affected**

- Defence operators
- Maintenance teams
- Mission planners and command units

**HUMS DATA**

that could predict failures weeks in advance remains under-analysed.

\$90B /year

US military spending on maintenance.

**The result?**

Lower readiness, higher costs, and avoidable downtime.

THE SOLUTION

AI-Powered Predictive Maintenance

Transforming asset data into early warnings and actionable insights for higher mission readiness.

KEY CAPABILITIES

**Predicts Failures Early**

Uses ML models to detect potential failures before they occur.

**Recommends Maintenance Actions**

Provides actionable recommendations with priority and timing.

**Monitors Asset Health in Real-Time**

Analyses sensor and operational data for continuous health assessment.

**Identifies Anomalies**

Detects abnormal behaviour in critical components.

**Supports Data-Driven Decisions**

Helps planners optimise maintenance schedules and resource allocation.

**Improves Operational Reliability**

Reduces unplanned downtime and extends asset lifespan.

KEY BENEFITS

**Reduced Downtime**

Minimise unexpected failures and operational disruptions.

**Lower Maintenance Costs**

Optimise maintenance schedules and resource usage.

**Higher Asset Availability**

Keep fleets and equipment ready for critical missions.

**Enhanced Safety**

Prevent catastrophic failures and ensure personnel safety.



OUR GOAL

**Keep critical assets mission-ready,
always.**

*"From data to decisions.
For a more resilient tomorrow."*



HOW IT WORKS

From Data to Decision

Transforming raw asset data into actionable maintenance intelligence.



EXAMPLE: ASSET HEALTH TREND



RECOMMENDED ACTION

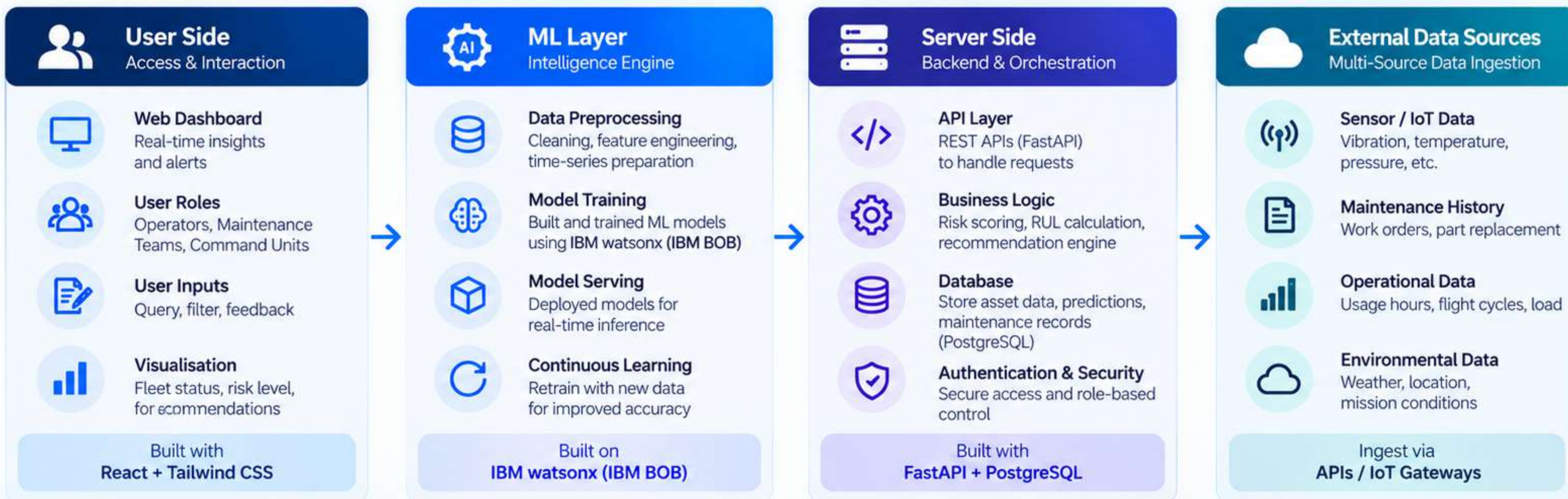
Inspect within 72 hours
to prevent potential failure.

SYSTEM ARCHITECTURE

Inside SentinelAI

A connected platform from data to decisions.

Integrating **data, AI** and actionable insights to keep critical assets **mission-ready**.



**END-TO-END
IMPACT**



Predict
Forecast failures early



Prevent
Reduce unplanned downtime



Protect
Ensure mission readiness

IBM TECHNOLOGY

Smarter Models. Real-World Impact.

We use IBM watsonx (BOB) to build, train and deploy our AI models for mission readiness.

IBM watsonx (BOB)

Powering our end-to-end model development.



Train
Models



Evaluate
& Optimize



Deploy
Models

OUR AI MODELS



Anomaly Detection

Detects unusual
patterns



RUL Prediction

Estimates remaining
useful life



Maintenance Recommendation

Suggests optimal
actions



Risk Assessment

Prioritizes high-
risk assets

A SMARTER TOMORROW
FOR CRITICAL ASSETS

OUR TEAM

Different Strengths. One Mission.

A passionate team working together to build a safer tomorrow.



Aayush Savaliya

Team Lead

- Complete working prototype
- Frontend development
- Backend development
- Deployment
- Project planning & coordination

*“From idea
to impact.”*



Jeel Pipaliya

ML Engineer

- Model development (4 models)
- Model evaluation & tuning
- Integration with application
- Research & experimentation

*“Turning
data into
intelligence.”*



Jevil Savani

Data & Model Analyst

- Data collection & preprocessing
- Model training support
- Performance analysis
- Insights & optimization

*“Better
data. Stronger
decisions.”*



Krish

QA & DevOps

- Testing and validation
- Maintain GitHub structure
- Bug fixes & performance checks
- Support deployment

*“Building
reliability
together.”*

IMPACT

Measurable Impact, Mission-Ready Outcomes

Turning predictive intelligence into real operational value.

**70%**

**Reduction in
Unexpected Failures**

Fewer in-mission disruptions

**30%**

**Lower Maintenance
Costs**

Optimised resource usage

**60%**

**Improvement in
Asset Uptime**

Higher operational availability

**85%**

Prediction Accuracy

Reliable and actionable insights



**Enhanced
Mission Readiness**

Safer operations,
greater confidence

BEYOND NUMBERS

**Proactive Operations**

Shift from reactive to
predictive maintenance.

**Resource Optimisation**

Better allocation of
personnel and spares.

**Extended Asset Life**

Maximise the value
of critical assets.

**Safety First**

Reduce risks for
equipment and personnel.

*“From data insights
to mission confidence.”*

SAFE ASSETS.
STRONGER TOMORROW.

Greater readiness.
Lower risk.
A stronger tomorrow.



PREDICT
Potential failures



PREVENT
Operational risks



PROTECT
Critical missions

A SMARTER TOMORROW
FOR CRITICAL ASSETS

THE ROAD AHEAD

Beyond the Hackathon

From a focused MVP to a scalable, enterprise-grade mission readiness platform.

*“Building today
for a more **resilient**
tomorrow.”*

01
TODAY



Predict

Asset health +
failure prediction

- HUMS data analysis
- Anomaly detection
- Risk assessment
- Real-time insights (MVP)

02
NEXT



Scale

More aircraft, vehicles
and equipment

- Expand to multiple platforms
- Handle larger data volumes
- Integrate diverse data sources
- Improve model accuracy

03
ADVANCE



Prescribe

RUL, digital twins and
recommended actions

- Advanced ML models
- Digital twin integration
- Optimised maintenance plans
- What-if scenario simulation

04
VISION



Mission Readiness

Enterprise-wide
predictive readiness

- Fleet-wide deployment
- Cross-domain integration
- Continuous learning & self-improvement
- Scalable, real-world impact

BEYOND ONE PLATFORM

One intelligence layer.
Multiple critical assets.



AIR



LAND



SEA



ENERGY

Thank You!

